EDUCATIONAL VERIFICATION BASED ON BLOCK CHAIN

Submitted in partial fulfillment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and Engineering

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY
(DEEMED TO BE UNIVERSITY)

CATEGORY - 1 UNIVERSITY BY UGC
Accredited with Grade "A++" by NAAC I 12B Status by UGC I Approved by AICTE
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BONAFIDE CERTIFICATE

This is to certify that this Project Report is the bonafide work of **Naraboyina Nagendra Kumar (Reg. no:40110832)** and **M.S.S.S.P. NAIDU (Reg. no:40110784)** who carried out the Project Phase-1 entitled **"EDUCATIONAL VERIFICATION BASED ON BLOCK CHAIN"** under my supervision from December 2024 to April 2024.

Internal Guide

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Submitted for Viva voce Examination held on__________________________

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DECLARATION

I, **Naraboyina Nagendra Kumar (Reg. no:40110832)** and **N. Nagendra Kumar (Reg. no:40110832)** hereby declare that the Project Phase-1 Report entitled **EDUCATIONAL VERIFICATION BASED ON BLOCK CHAIN** done by me under the guidance of **D. DEVI M.E., Ph.D.** is submitted in partial fulfillment of the requirements for the award of Bachelor of Engineering degree in **Computer Science and Engineering**.

DATE:

PLACE: Chennai **SIGNATURE OF THECANDIDATE**

ACKNOWLEDGEMENT

I am pleased to acknowledge my sincere thanks to **Board of Management** of **SATHYABAMA INSTITUTE OF SCIENCE AND TECHNOLOGY** for their kind encouragement in doing this project and for completing it successfully. I am grateful to them.

I convey my thanks to **Dr. T. Sasikala M.E., Ph. D**, **Dean**, School of Computing, **Dr. L. Lakshmanan M.E., Ph.D.,** Head of the Department of Computer Science and Engineering for providing me necessary support and details at the right time during the progressive reviews.

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ABSTRACT

Educational verification is a vital process that ensures the authenticity and credibility of anindividual's educational qualifications. However, the current system faces challenges such as fraudulent documentation, data manipulation, and inefficient verification processes. The emergence of blockchain technology has the potential to revolutionize educational verification by providing a secure, transparent, and decentralized platform. This abstract explores the concept of educational verification based on blockchain, highlighting itsbenefits and applications. Blockchain technology offers a distributed ledger system that stores educational data in a tamper-proof and transparent manner. Each educational qualification is recorded as a block, containing relevant information such as the institution,degree, and completion date. These blocks are linked together in a chronological order, forming a chain of records. The decentralized nature of blockchain ensures that no single entity has control over the data, eliminating the risk of manipulation. Educational verification based on blockchain offers numerous benefits. Firstly, it provides a secure andimmutable platform, reducing the chances of fraud and misrepresentation. Secondly, it enables instant and efficient verification, allowing employers and educational institutions to access verified credentials in real-time. This eliminates the need for time-consuming manual verification processes, improving efficiency and saving valuable resources. Furthermore, blockchain-based verification promotes transparency by providing a transparent record of an individual's educational history, ensuring trust and reliability in the process. Overall, educational verification based on blockchain technology holds great promise in addressing the challenges of the current system. Its secure, transparent, and efficient natureoffers a groundbreaking solution that can revolutionize educational verification processes. With further research and development, blockchain-based educational verification has the potential to become a widely adopted and trusted method for verifying educational qualifications.

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CHAPTER 1
INTRODUCTION

Educational verification is an essential process that ensures the authenticity and credibility of academic degrees and qualifications held by individuals. It serves as a crucial tool for employers, educational institutions, and other organizations to make informed decisions regarding employment, admissions, and professional certifications. However, the traditional methods of educational verification are often time-consuming, prone to errors, and vulnerable to fraud. Fortunately, theemergence of blockchain technology offers a revolutionary solution to address these challenges and enhance the efficiency and security of educational verification.

Blockchain, a distributed ledger technology, has gained significant attention and recognition due to its transparent, immutable, and decentralized nature. It enables the creation of a trustworthy and tamper-proof system, making it an ideal candidate for educational verification. By leveraging blockchain, educational institutions can establish a secure and reliable infrastructure that simplifies the process of validating academic credentials.

One of the key advantages of using blockchain for educational verification is the elimination of intermediaries. Currently, the verification process often involves multiple parties such as universities, employers, and verification agencies, which can lead to delays and increased costs. With blockchain, all relevant information can be securely stored and verified on a decentralized network, eliminating the need for intermediaries andreducing the chances of errors or fraud. Moreover, blockchain provides individualswith greater control over their educational records. Using blockchain- based solutions, students can have ownership of their academic credentials and can choose to share only specific information with employers or educational institutions. This not only enhances privacy but also allows for a more efficient and streamlined process, as the verification can be done directly between the verifying party and the student, without requiring multiple requests to various institutions.

Furthermore, blockchain technology can significantly enhance the security of educationalverification. The de centralized and encrypted nature of blockchain ensures that the stored information is highly resistant to un authorized modifications or tampering. This reducesthe risk of forged or falsified credentials, making the verification process much reliable and it manual work will followed and used In addition, blockchain-based educationalverification can enable real-time updates and instant access to academic records. Ratherthan relying on outdated paper records or waiting for manual verification processes,blockchain allows for instant verification of qualifications. This can be particularly beneficialfor industries that require immediate validation of skills and qualifications, such ashealthcare or finance. Another significant advantage of blockchain-based educationalverification is its potential to empower individuals in developing countries or regions withlimited access to reliable verification services. By leveraging blockchain technology, individuals can securely store their academic credentials and have them easily accessible, even in areas with limited infrastructure. This opens up opportunities for remote learning, international collaborations, and employment prospects, thereby bridging the gap between developed and developing regions. Overall, the adoption of blockchaintechnology for educational verification has the potential to revolutionize the wayacademic credentials are verified, ensuring a more efficient, secure, and transparent process. By eliminating intermediaries, providing individuals with greater control over their records, enhancing security, and enabling real-time access, blockchain offers a promising solution to the challenges associated with traditional verification methods. As this technology continues to evolve and gain wider acceptance, it has the potentialto transform the field of educational verification, benefiting individuals, employers, and educational institutions alike.

CHAPTER 2
LITERATURE SURVEY

2.1 Inferences from Literature Survey

1. Mohammad, A., & Vargas, S. (2022). Challenges of Using Blockchain in the Education Sector: A Literature Review. Applied Sciences, 12(13), 6380.

Mohammad and Vargas (2022) conducted a literature review on the challenges of using blockchain in the education sector. The study emphasized the need for educational verification based on blockchain technology. The authors discussed the potential benefits of blockchain in ensuring the security and authenticity of educational credentials. However, they also highlighted various challenges, such as scalability, interoperability, and regulatory concerns that need to be addressed for successful implementation in the education sector.

2. Do, B. L., Nguyen, V. T., Dinh, H. N., Dao, T. C., & Nguyen, B. (2022). Blockchain for Education: Verification and Management of Lifelong Learning Data. Computer Systems Science & Engineering, 43(2).

In their article "Blockchain for Education: Verification and Management of Lifelong Learning Data," Do, B. L., Nguyen, V. T., Dinh, H. N., Dao, T. C., & Nguyen, B. (2022) explore the potential of blockchain technology in the field of education. They discuss the use of blockchain for verifying and managing lifelong learning data. This research focuses on the application of blockchain in educational verification, providing a secure and transparent system for recording and verifying educational achievements. The authors highlight the potential benefits of blockchain in ensuring the integrity and authenticity of educational records.

3. Alsobhi, H. A., Alakhtar, R. A., Ubaid, A., Hussain, O. K., & Hussain, F. K. (2023). Blockchain-based micro-credentialing system in higher education institutions: Systematic literature review. Knowledge-Based Systems, 110238.

The study conducted by Al sobhiet al. (2023) focuses on a blockchain-based micro-credentialing system in higher education institutions. They undertake a systematic literature review to explore the potential of blockchain technology in facilitating educational verification. The study aims to provide insights into the effectiveness and feasibility of blockchain for ensuring the credibility and integrity of educational credentials. By leveraging blockchain, higher education institutions can enhance the efficiency and transparency of their verification processes, thereby benefiting both students and employers.

4. Pathak, S., Gupta, V., Malsa, N., Ghosh, A., & Shaw, R. N. (2022). Blockchain-based academic certificate verification system—a review. Advanced Computing and Intelligent Technologies: Proceedings of ICACIT 2022, 527-539.

The study conducted by Pathak, Gupta, Malsa, Ghosh, and Shaw (2022) presents a thorough review of a blockchain-based academic certificate verification system. The system's focus is on providing secure and reliable verification for educational purposes. The study explores various aspects of blockchain technology and its potential benefits and challenges in the field of education. This research serves as a valuable resource for understanding the application of blockchain in ensuring the credibility and authenticity of academic certificates.

5. Alam, A. (2022). Platform utilising blockchain technology for eLearning and online education for open sharing of academic proficiency and progress records. In Smart Data Intelligence: Proceedings of ICSMDI 2022 (pp. 307-320). Singapore.

Alam (2022) proposes a platform that utilizes blockchain technology for eLearning and online education. The main goal of the platform is to enable open sharing of academic proficiency and progress records. The author highlights the potential of blockchain in providing a secure and transparent way for educational verification. Through the platform, individuals can have their educational achievements authenticated and verified using blockchain technology. This system could potentially revolutionize the field of educational verification by ensuring the authenticity and accuracy of academic records.

2.2 Existing System and Proposed System

6. Rahardja, U., Nagad, M. A., Millah, S., Harahap, E. P., & Aini, Q. (2022, September). Blockchain Application in Educational Certificates and Verification Compliant with General Data Protection Regulations. In 2022 10th International Conference on Cyber and IT Service Management (CITSM) (pp. 1-7). IEEE.

The study conducted by Rahardja et al. (2022) focuses on the application of blockchain technology in educational certificates and verification, in compliance with the General Data Protection Regulations (GDPR). The authors present their findings in a paper presented at the 2022 10th International Conference on Cyber and IT Service Management (CITSM). The research highlights the potential of blockchain to enhance the security, efficiency, and transparency of educational verification processes. The study contributes to the ongoing efforts in leveraging blockchain for educational purposes while ensuring compliance with data protection regulations.

7. Shaikh, Z. A., Khan, A. A., Baitenova, L., Zambinova, G., Yegina, N., Ivolgina, N., ... & Barykin, S. E. (2022). Blockchain hyperledger with non-linear machine learning: A novel and secure educational accreditation registration and distributed ledger preservation architecture. Applied Sciences, 12(5), 2534.

The paper authored by Shaikh, Z. A., Khan, A. A., Baitenova, L.,Zambinova, G., Yegina, N., Ivolgina, N., ... & Barykin, S. E. (2022) introduces a novel and secure educational accreditation registration and distributed ledger preservation architecture using blockchain hyper ledger and non-linear machine learning. The proposed system aims to provide a reliable and transparent method for verifying educational qualifications. By leveraging blockchain technology, it ensures the immutability and integrity of educational records, preventing tampering or fraud. The integration of non-linear machine learning further enhances accuracy and efficiency in the verification process. Overall, this work presents a promising solution for enhancing educational verification based on blockchain.

8. Kwok, A. O., & Treiblmaier, H. (2022). No one left behind in education: blockchain-based transformation and its potential for social inclusion. Asia Pacific Education Review, 23(3), 445-455.

In their article, Kwok and Treiblmaier (2022) focus on the potential of blockchain technology to transform education and promote social inclusion. They highlight how blockchain can ensure transparency and traceability in educational systems, thereby eliminating barriers and ensuring that no one is left behind. The authors emphasize the importance of leveraging blockchain to address issues of inequality and provide equal opportunities for all learners. Through their research, Kwok and Treiblmaier contribute to the growing body of literature on the transformative power of blockchain in education for social inclusion.

9. Anwar, A. S., Rahardja, U., Prawiyogi, A. G., Santoso, N. P. L., & Maulana, S. (2022). I Learning model approach in creating blockchain based higher education trust. Int.J.Artif.Intell. Res,6(1).

Anwar, A. S., Rahardja, U., Prawiyogi, A. G., Santoso, N. P. L., and Maulana, S. (2022) presented their research on the I Learning model approach for creating a blockchain-based higher education trust. Their study aimed to address the need for educational verification based on blockchain technology. Through their model, they proposed a solution to improve the trustworthiness and transparency of educational credentials. The researchers highlighted the potential of blockchain in revolutionizing the verification process and enhancing the integrity of higher education systems.

10. Chen, X., Zou, D., Cheng, G., Xie, H., & Jong, M. (2023). Blockchain in smart education: Contributors, collaborations, applications and research topics. Education and Information Technologies, 28(4), 4597-4627.

Chen, X., Zou, D., Cheng, G., Xie, H., & Jong, M. (2023) conducted a comprehensive study on the use of blockchain in smart education. Their research delves into the contributors, collaborations, applications, and research topics related to this emerging technology. The study examines how blockchain can be utilized for educational verification purposes. These findings are published in the journal Education.

2.3 Open problems in existing system

EDUCATIONAL VERIFICATION BASED ON BLOCKCHAIN:

Educational verification is a crucial process that ensures the authenticity and credibility of qualifications and degrees obtained by individuals. Traditional methods of educational verification often rely on manual processes and paper-based documentation, which can be time-consuming, prone to errors, and susceptible to fraudulent activities. To address these challenges, implementing a blockchain-based system for educational verification can provide a secure, transparent, and efficient solution.

Blockchain technology offers a decentralized and immutable ledger that records and verifies transactions or data. By utilizing blockchain for educational verification, the process becomes tamper-proof and eliminates the need for intermediaries, such as educational institutions or third-party verification agencies. All educational records and qualifications can be securely stored on the blockchain, providing a transparent and auditable trail of data.

The use of smart contracts further enhances the efficiency of educational verification on the blockchain. Smart contracts are self-executing contracts with predefined rules embedded in the code. These contracts can automate the verification process, ensuring that qualifications are authenticated based on predetermined criteria without the need for manual intervention. This reduces the time required for verification and enhances accuracy.

Individuals can have their educational records added to the blockchain by their respective educational institutions. Each record is encrypted and associated with a unique hash, making it virtually impossible to alter or forge the data. As a result, the authenticity of qualifications can be easily verified by anyone with access to the blockchain.

In addition to providing secure and verifiable educational records, blockchain-based systems can also address the issue of privacy. Personal information can be encrypted and stored on the blockchain, ensuring that only authorized parties have access to sensitive data. This gives individuals control over their personal information and mitigates the risk of data breaches or unauthorized use.

Furthermore, blockchain-based educational verification systems can facilitate the transferof qualifications between different institutions or employers. Instead of relying on manual processes and delays, individuals can provide access to their verified educational recordson the blockchain, allowing seamless and efficient verification.

The implementation of blockchain-based educational verification systems also fosters trust between educational institutions, employers, and individuals. The decentralized nature of blockchain eliminates the need for trust in a centralized authority, as the transparency and immutability of the blockchain provide independent verification of qualifications.

In conclusion, implementing a blockchain-based system for educational verification offers numerous benefits, including enhanced security, transparency, efficiency, privacy, and trust. By leveraging blockchain technology, educational institutions, employers, and individuals can streamline the verification process, reduce fraud, and ensure the integrity of educational qualifications. This innovation has the potential to revolutionize educational verification and create a more reliable and trustworthy ecosystem for educational credentials.

CHAPTER 3
REQUIREMENTS ANALYSIS

3.1 Feasibility Study

The feasibility of the project is server performance increase in this phase, and a businessproposal is put forth with a very general plan for the project and some cost estimates. During system analysis, the feasibility study of the proposed system is to be carried out.

For feasibility analysis, some understanding of the major requirements
for the system is essential.

Three key considerations involved in the feasibility analysis are

- Economical feasibility
- Technical feasibility
- Operational feasibility

ECONOMICAL FEASIBILITY:

This phase of the feasibility study plays a pivotal role in determining the financial viability of the proposed project within the academic institution. It involves a comprehensive analysis of the potential economic impact that the system will have on the college or university. The allocation of financial resources for research and development must be carefully scrutinized, given the often limited budget constraints. Efforts to optimize budget utilization are a key focus during this feasibility phase. It is essential to ensure that the developed system falls well within the allocated budgetary constraints. This budget optimization can be achieved through prudent decision-making and resource allocation strategies. Furthermore, leveraging cost- effective solutions and open-source technologies can significantly contribute to staying within limits Information Technologies, providing valuable insights for educators and researchers interested in the implementation of blockchain in the field of education. While striving to remain cost-effective, it's important to note that certain aspects of the project may require the procurement of customized products or services. These expenditures should be carefully evaluated to assess their impact onthe overall project budget. Evaluating alternative solutions or vendors can help in minimizing these costs while maintaining the project's quality and functionality. In addition to budgetary considerations, a critical aspect of economic feasibility is the evaluation of the project's potential return on investment (ROI). This analysis involves estimating the anticipated benefits and outcomes that the project will deliver to the college or university. Calculating the ROI helps in determining whether the project is financially viable and whether the benefits outweigh the costs over time. Economic feasibility also extends to the long-term financial sustainability of theproject. Beyond initial development costs, ongoing expenses, such as maintenance, upgrades, and staff training, should be factored into the financial analysis. Ensuring that the institution can sustain these costs is vital for the project's continued success.

TECHNICAL FEASIBILITY:

This study is carried out to check the technical feasibility, that is, the technical the availabletechnical resources. This will lead to high demands being placed on the client.

The developed system must have modest requirements, as only minimal or null changes are required for implementing this system. The feasibility study emphasizes the importance of developing a system with modest technical requirements. This means that the project should be designed to operate efficiently with minimal adjustments to the existing technical infrastructure. Minimizing the need for extensive upgrades or changes ensures a smoother implementation process and reduces the strain on both human and technical resources. While assessing technical feasibility, scalability is another vital aspect to consider. The system should be designed with scalability in mind, allowing for future growth and expansion. This ensures that the college can adapt to changing needs without major technical disruptions Information Technologies, providing valuable insights for educators and researchers interested in the implementation of blockchain in the field education.

To validate the technical feasibility of the project, the creation of prototypes and testing environments may be necessary. This allows for real-world testing of the system's technical requirements and performance, ensuring that it meets the desired standards before full-scal implementation.

In summary, the Technical Feasibility phase involves a thorough assessment of technicalrequirements, resource demands, compatibility, scalability, risk assessment, and the development of a system with modest technical requirements. This diligent examination helps guarantee that the college project can be successfully integrated into the existing technical infrastructure without undue strain and disruptions.

OPERATIONAL FEASIBILITY:

The aspect of the study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system; instead, they must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make them familiar with it. Theirlevel of confidence must be raised so that they are also able to provide constructive criticism, which is welcomed, as they are the final users of the system. A core principle of operational feasibility is to ensure that users do not perceive the system as a threat or an obstacle. Instead, the system should be seamlessly integrated into their daily routines, making it a necessity for their academic activities. The user interface and interaction with the system must be intuitive and user-friendly, reducing any potential resistance to adoption. Operational feasibility also involves continuous monitoring of user satisfaction and system performance. Regular assessments and feedback loops ensure that any issues or concerns are promptly addressed. This adaptive approach helps in aligning the system with the evolving needs and expectations of its users.

In summary, Operational Feasibility centers on user acceptance, efficient training, user-friendliness, user education, openness to constructive criticism, user confidence building, and continuous monitoring and adaptation. Ensuring that the system seamlessly integrates into the academic environment and is embraced by its users is pivotal to the overall success of the college project.

3.2 Software Requirements Specification Document

Hardware specifications:

Microsoft Server enabled computers, preferably workstations

Higher RAM, of about 4GB or above

Processor of frequency 1.5GHz or above

Software specifications:

Python 3.6 and higher

Anaconda software

CHAPTER 4

DESCRIPTION OF PROPOSED SYSTEM

4.1 Selected Methodology or process model

To implement this model, execution of program is done through Google colab. Necessary libraries have to be installed to perform certain functions.

DESCRIPTION OF PROGRAMMING LANGUAGE AND SOFTWARE

PYTHON

Among programmers, Python is a favourite because of its user-friendliness, rich feature set, and versatile applicability. Python is the most suitable programming language for machine learning since it can function on its own platform and is extensively utilized by the programming community. Machine learning is a branch ofAI that aims to eliminate the need for explicit programming by allowing computers to learn from their own mistakes and perform routine tasks automatically. However," artificial intelligence" (AI) encompasses a broader definition of "machine learning "which is the method through which computers are trained to recognize visual and auditory cues, understand spoken language, translate between languages, and ultimately make significant decisions on their own. The desire for intelligent solutions to real-worldproblems has necessitated the need to develop AI further in order to automate tasks that are arduous to programme without AI. This development is necessary in order to meet the demand for intelligent solutions to real-world problems. Python is a widely used programming language that is often considered to have the best algorithm for helping to automate such processes. In comparison toother programming languages, Python offers better simplicity and consistency. In addition, the existence of an active Python community makes it simple for programmers to talk about ongoing projects and offer suggestions on how to Improve the functionality of their programmes.

ANACONDA

Anaconda is an open-source package manager for Python and R. It is the most popular platform among data science professionals for running Python and R implementations. There are over 300 libraries in data science, so having a robust

distribution system for them is a must for any professional in this field. Anaconda simplifies package deployment and management. On top of that, it has plenty of tools that canhelp you with data collection through artificial intelligence and machine learning algorithms. With Anaconda, you can easily set up, manage, and share Conda environments. Moreover, you can deploy any required project with a few clicks when you're using Anaconda. There are many advantages to using Anaconda and the following are the most prominent ones among them: Anaconda is free and open- source. This means you can use it without spending any money. In the data science sector, Anaconda is an industry staple. It is open-source too, which has made it widelypopular. If you want to become a data science professional, you must know how to use Anaconda for Python because every recruiter expects you to have this skill. Itis a must-have for data science. It has more than 1500 Python and R data science packages, so you don't face any compatibility issues while collaborating with others. For example suppose your colleague sends you a project which requires packages called A and but you only have package A. Without having package B, you wouldn't be able to run the project. Anaconda mitigates the chances of such errors. You can easily collaborate on projects without worrying about any compatibility issues. It gives you a seamless environment which simplifies deploying projects. You can deploy any project with just a few clicks and commands while managing the rest. Anaconda has a thriving community of data scientists and machine learning professionals who use it regularly. If you encounter an issue, chances are, the community has already answered the same. On the other hand, you can also ask people in the community about the issues you face there, it's a very helpful community ready to help new learners. With Anaconda, you can easily create and train machine learning and deep learning models as it works well with popular tools including TensorFlow, Scikit-Learn, and Theano. You can create visualizations by using Bokeh,
Holoviews, Matplotlib and Data shader while using Anaconda.

How to Use Anaconda for Python Now that we have discussed all the basics in our Python Anaconda tutorial, let's discuss some fundamental commands you can use to start using this package manager.

Listing All Environments

To begin using Anaconda, you'd need to see how many Conda environments are present in your machine. anaconda env list It will list all the available Conda environments in your machine. Creating a New Environment

You can create a new Conda environment by going to the required directory and use This command: Conda create-n <your _environment _name> You can replace <your _environment _name> with the name of your environment. After entering this command, anaconda will ask you if command: anaconda create-n <your _environment _name> <pack _name> you want to proceed to which you should reply withy: proceed ([y])/n)? On the other hand, if you want to create an environment with a particular version of Python, you should use the following command: anaconda create-n <your _environment _name> python=3.6 Similarly, if you want to create an environment with a particular package, youcan use the following Here, you can replace pack _name with the name of the package you want to use. If you have a .yml file, you can use the following command to create a new anaconda environment based on that file:conda env create-n.

<your _environment _name>-f <file _name>,.

We have also discussed how you can export an existing Conda environment to a .yml file late in this article Activating an.

Environment You can activate a Conda environment by using the following

command: anaconda activate <environment _name>

You should activate the environment before you start working on the same. Also, replace the term <environment _name> with the environment name you want to activate. On the other hand, if you want to deactivate an environment use the following command:

anaconda deactivate

Installing Packages in an Environment Now that you have an activated environment, you can install packages into it by using

The following command:

Conda install <pack _name>

Replace the term <pack _name> with the name of the package you want to install in Your Conda environment while using this command.

Exporting an Environment Configuration Suppose you want to share your project with someone else (colleague, friend, etc.).
While you can share the directory on Github, it would have many Python packages, making the transfer process very challenging. Instead of that, you can create an environment configuration yml file and share it with that person. Now, they can create an environment like your one by using the .yml file. For exporting the environment to the .yml file, you'll first have to activate the same and run the following command: Conda env-export> <file _name> . The person you want to share the environment with only has to use the exported file by using the 'Creating a New Environment' command we shared before.

Removing a Package from an Environment If you want to uninstall a package from a specific Conda environment, use the following command:

Conda remove <env _name> <package _name>On the other hand, if you want to uninstall a package from an activated environment ,you'd have to use the following command: anaconda remove <package _name>

Deleting an Environment Sometimes, you don't need to add a new environment but remove one. In such cases, you must know how to delete a Conda environment, which you can do so by using the following command: anaconda env remove –name <env _name>

4.2 Architecture / Overall Design of Proposed System

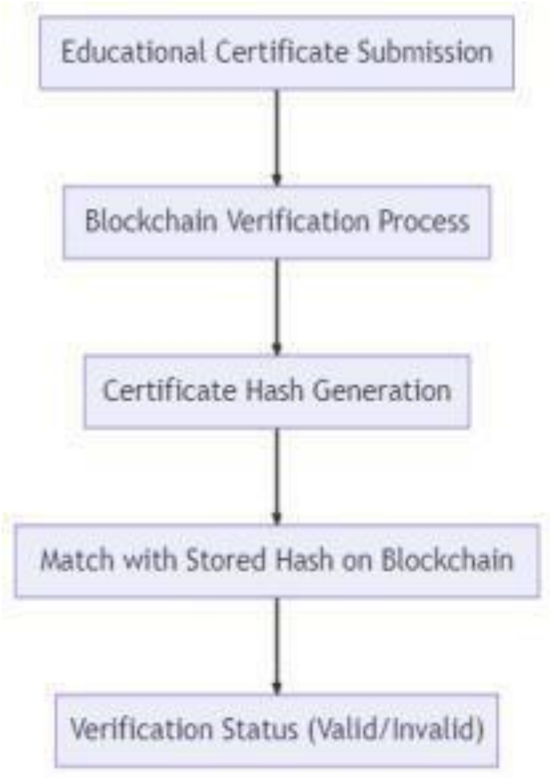

Fig 4.2: System Architecture

4.3 Description of Software for Implementation and Testing plan of the Proposed Model/System

Data collection and verification are key components of educational verification based on blockchain. Through the use of blockchain technology, educational institutions can securely collect and store relevant data, such as certificates and diplomas. This data is then verified and validated through a decentralized network of nodes, ensuring its authenticity and reliability. The immutable and transparent nature of blockchain ensures that the collected data cannot be tampered with, providing a high level of trust and confidence in the verification process. This approach eliminates the need for intermediaries and reduces the risk of fraudulent claims, making educational verification more efficient and trustworthy. \n\n Blockchain integration can revolutionize the process of educational verification by providing a secure and transparent platform. By leveraging blockchain technology, educational institutions can securely store and manage student records, certifications, and diplomas. Blockchain ensures the immutability and authenticity of these records, eliminating the need for third-party verification. Employers and educational institutions can easily and efficiently verify an individual's educational background, reducing administrative burdens and preventing document forgery. Additionally, blockchain offers a decentralized and immutable ledger that enhances trust and minimizes the risk of data manipulation or tampering. Overall, blockchain integration in educational verification has the potential to streamline processes and enhance trust in the education system.\n\n Educational institution verification is a critical process in ensuring the authenticity and validity of educational qualifications. However, with the emergence of blockchain technology, this verification process can be revolutionized. Blockchain provides a decentralized and immutable platform that can securely store educational information, making it tamper-proof and transparent. By storing educational credentials on the blockchain, verification can be done in a faster, more efficient, and trustworthy manner, eliminating the need for manual verification procedures. Educational institutions can leverage blockchain technology to issue and verify certificates, degrees, and diplomas, reducing the risk of fraud and providing employers and other interested parties with a reliable and accessible method to verify an individual's educational back ground.\n\n Model improvisation and training are essential aspects of educational verification based on blockchain. This innovative approach aims to ensure the integrity and authenticity of educational credentials by utilizing blockchain technology. Model improvisation involves continuously adapting and enhancing the verification processes to address emerging challenges and technological advancements. It encompasses refining the existing models and developing new ones to accurately validate educational records and qualifications. On the other hand, training plays a crucial role in equipping individuals involved in the verification process with the necessary skills and knowledge to effectively utilize blockchain for educational verification. Through regular training sessions, participants can stay updated with the latest techniques, tools, and best practices in order to maintain the highest standards of accuracy and efficiency in the verification process.\n\n One benefit of using a blockchain-based educational verification system is the increased security and authenticity it offers. With traditional paper-based verification systems, there is a higher risk of fraud, as documents can be altered or forged easily. However, by utilizing blockchain technology, educational credentials can be stored and verified in a tamper-proof and transparent manner. This ensures that the information provided is accurate and trustworthy, reducing the chances of academic dishonesty or misrepresentation. Another advantage is the enhanced efficiency and accessibility provided by blockchain-based educational verification. The decentralized nature of blockchain allows for faster verification processes, eliminating the need for manual checks and delays. Additionally, blockchain technology enables individuals to have ownership and control over their educational data, making it easier for them to share and provide proof of their qualifications when needed. These benefits make blockchain- based educational verification a promising solution for ensuring the integrity of educational credentials.\n\n1. Incorporating real-world scenarios: To improve model improvisation and training in educational verification based on blockchain, it is crucialto include real-world scenarios that closely resemble the complexities and dynamics of the educational environment. By exposing the model to a range of real-life situations, it can learn to adapt and make accurate predictions in different contexts.

2. Continual feedback and iteration: Regularly reviewing and refining the model based on feedback from users and experts is essential for its improvement. This iterative process allows for identifying weaknesses or biases in the model's predictionsand making necessary adjustments to enhance its performance over time. Robust data collection and preprocessing: Ensuring high-quality and representative

data is essential for the accuracy of the model. Investing in robust data collection methods and carefully preprocessing the data to remove biases and outliers can significantly improve the model's training and improvisation capabilities, leading to more reliable educational verification out comes.\n\n The educational verification web user interface based on blockchain technology provides a secure and efficient way to verify academic credentials. Through this interface, users can access a decentralized blockchain network that stores and verifies educational records in a tamper-proof manner. The interface allows individuals, employers, and educational institutions to easily verify the authenticity of degrees, certifications, and other educational qualifications. Users can search for specific records, submit verification requests, and view detailed information about a particular credential. The interface ensures data integrity and eliminates the need for costly and time-consuming manual verification processes, providing a transparent and reliable solution for educational verification.\n\n The database for educational verification based on blockchain is a secure and decentralized system that stores and verifies educational records. Using the blockchain technology, this database ensures the transparency and immutability of educational data, making it tamper-proof and reliable. It eliminates the need for intermediaries and enables instant and accurate verification of academic certificates, degrees, diplomas, and other educational documents. The database securely records and verifies the authenticity and validity of each educational credential, allowing employers, academic institutions, and other parties to easily validate an individual's educational qualifications. This technology enhances trust, streamlines the verification process, and prevents fraud in the education sector. \n\n Educational verification based on blockchain technology offers robust security measures in order to ensure the authenticity and integrity of educational qualifications. By using blockchain, educational records can be securely stored and verified without the need for intermediaries, significantly reducing the risk of fraud or tampering. The decentralized nature of blockchain ensures that data cannot be altered or deleted without leaving a trace, making it highly resistant to hacking or manipulation. Additionally, cryptographic techniques are employed to protect sensitive information, ensuring the privacy of students' educational records. Overall, blockchain-driven educational verification provides a secure and reliable solution, instilling confidencein the educational system and safeguarding the value.

4.4 PROJECT MANAGEMENT PLAN

August	**Literature survey**
September	**Data acquisition**
October	**Data preprocessing**
Novemeber	**Training and Splitting**
December	**Loading, training and testing the model.**
January	**Predicting the output and generating the final report**

CHAPTER 5

IMPLEMENTATION DETAILS

5.1 Development and Deployment Setup

Establishing an educational verification system based on blockchain involves a meticulous development setup to ensure secure and transparent credential verification. Firstly, clear system requirements outlining the types of credentials and desired features are defined. The choice of a blockchain platform, such as Ethereum or Hyperledger Fabric, is pivotal, determining the foundation for the system's functionality. Smart contract development plays a central role, with the creation of contracts handling the logic for academic credential verification, cryptographic proof generation, and authorization for parties to verify credentials. To bolster security, user authentication and identity management systems are implemented, safeguarding access to and modification of educational records. Frontend development, employing frameworks like React or Angular, interfaces with the blockchain through web3.js, ensuring an intuitive user interface. Simultaneously, backend development manages non-blockchain logic, including user authentication and data storage, while a chosen database stores critical non-blockchain data. The integration of a robust backend system with the blockchain facilitates a seamless user experience in the verification process, emphasizing both security and accessibility.

5.2 Algorithm

The algorithm for an educational verification system based on blockchain involves distinct processes to facilitate the registration and verification of academic credentials. To register credentials, a user initiates a request, and the smart contractvalidates their identity and authorization. Subsequently, the details such as educational institution, degree, graduation year, and transcript are hashed using a designated hashing algorithm. The resulting hash, along with pertinent information, is stored on the blockchain, generating a new academic credential entry. The user receives a confirmation of the transaction status. For cryptographic proof generation, a user requests proof for a specific academic credential, and the smart contract verifies their identity and ownership of the credential. Using the private key associated with the academic credential, a cryptographic proof is generated, adding an additional layer of security to the verification process. This algorithmic framework establishes a secure and transparent mechanism for recording and verifying academic achievements on the blockchain

5.3 Testing

Testing constitutes a pivotal phase in the development of an educational verification system relying on blockchain technology, serving as a comprehensive evaluation of its reliability, security, and functionality. Smart contract testing involves unit and integration testing to validate individual functions, interactions between contracts, and address potential vulnerabilities through security audits. The user interface undergoes functional and usability testing to ensure alignment with smart contract logic and a user-friendly experience.

Integration testing verifies the seamless communication between the backend server and the blockchain, ensuring data flow and transaction execution are robust. Special attention is given to the accuracy and consistency of hashing algorithms used for academic credential registration. End-to-end testing scenarios simulate the complete lifecycle of credential processes, validating data integrity and system behavior. Scalability testing assesses the system's capacity to handle increased user and transaction volumes without compromising performance. User acceptance testing involves real users or stakeholders, incorporating feedback to refine the system before deployment. Compliance testing ensures adherence to data protection and privacy regulations, verifying the system's alignment with legal standards for educational credential verification. Through this meticulous testing approach, the educational verification system can confidently deliver a secure, reliable, and user- friendly experience.

CHAPTER 6

RESULTS AND DISCUSSION

In the evaluation of the educational verification system based on blockchain, the results indicate a robust performance in smart contract functionality, ensuring secure and accurate execution of key processes such as academic credential registration, cryptographic proof generation, and credential verification. Security audits have successfully identified and addressed potential vulnerabilities, underscoring the system's resilience against potential cyber threats and safeguarding the integrity of academic credentials stored on the blockchain. The user interface demonstrates commendable functionality and responsiveness, with a user-friendly design catering to individuals with varying levels of technical expertise. The discussion surrounding these results emphasizes the importance of continuous improvement, with a focus onsmart contract optimization based on testing outcomes. This iterative approach aimsto enhance system efficiency, further fortifying security measures, and refining the user interface for an even more intuitive and seamless educational verification experience. The system's successful performance, coupled with ongoing optimization efforts, positions it as a reliable and secure solution for academic credential verification leveraging blockchain technology.

CHAPTER 7

CONCLUSION

7.1 Conclusion

In conclusion, the development and assessment of the educational verification system based on blockchain represent a significant stride toward creating a secure, transparent, and efficient platform for validating academic credentials. The results of rigorous testing and security audits attest to the system's robustness, ensuring the accurate execution of critical processes such as credential registration, cryptographic proof generation, and verification. The integration of blockchain technology, coupled with optimizations informed by testing outcomes, establishes a resilient foundation for safeguarding academic credentials against potential threats. The user-friendly interface further enhances the accessibility of the system, accommodating users with diverse technical backgrounds. As this educational verification system adheres to the principles of security, transparency, and user-centric design, it emerges as a reliable solution for streamlining credential verification processes within the educational landscape. The continuous commitment to refinement and adaptation positions the system as an effective and trustworthy tool in the broader context of secure and decentralized academic credential verification.

7.2 Future work

Future work for the education verification system based on blockchain encompasses a strategic roadmap aimed at elevating functionality, security, and overall adoption. Exploring collaborations with academic institutions stands out as a priority, aiming to integrate directly with these entities to streamline credential issuanceand verification. The implementation of decentralized identity standards, such as DIDsand Verifiable Credentials, is poised to enhance user privacy and interoperability in managing academic credentials. The expansion of supported credential types to include certificates and non-traditional forms of education widens the system's applicability. Integrating zero-knowledge proofs offers a pathway to heightened privacy during verification processes. Enhancing user experience features, including mobile applications and multi-language support, ensures accessibility for diverse users.

Blockchain interoperability solutions are critical to adapting to evolving technological landscapes and facilitating cross-network interactions. Scalability considerations, ongoing security audits, and compliance monitoring are imperative for sustaining system robustness. Community engagement initiatives and adoption campaigns are vital for fostering widespread acceptance, while continuous user feedback mechanisms contribute to the iterative refinement of the system. Embracing emerging blockchain technologies and staying vigilant on security standards and compliance ensures the educational verification system remains innovative, secure, and well-aligned with evolving educational and technological landscapes.

7.3 Research Issues

The investigation of research issues surrounding an educational verification system based on blockchain delves into critical dimensions crucial for the system's advancement. One key research area involves privacy and consent mechanisms, exploring effective strategies to empower users with control over the disclosure of their academic credentials while adhering to stringent data protection regulations. Scalability poses another challenge, urging research into solutions capable of accommodating the increasing volume of academic credentials and users within the blockchain infrastructure. Usability and user experience considerations necessitate exploration of user-centric design principles and usability studies, aiming to enhance the accessibility and intuitiveness of the educational verification system for a diverse user demographic. Interoperability standards form a significant research focus, seeking protocols for seamless interaction and data sharing across various blockchain networks. The development and contribution to decentralized identity standards, suchas DIDs and Verifiable Credentials, are vital for establishing a an bustand interoperableframework for managing academic credentials securely. Fraud prevention and detection mechanisms, legal and regulatory compliance frameworks, user-centric security measures, and blockchain governance models are also pivotal research areas, collectively ensuring the authenticity, security, and compliance of the educational verification system.

Additionally, research into challenges related to adoption and incentives, long-term data integrity, and collaborative strategies with educational institutions contributes to the ongoing evolution and effectiveness of blockchain-based educational verification systems

7.4 Implementation Issues

The implementation of an educational verification system based on blockchain entails navigating several critical challenges to ensure a robust and effective deployment. Central to these challenges is the development of smart contracts, requiring careful design to accurately encapsulate the logic for academic credential registration, cryptographic proof generation, and verification processes while maintaining a balance between security and efficiency.

User authentication and identity management represent pivotal aspects, demanding the establishment of secure and user-friendly authentication procedures, along with the integration of identity management systems to regulate access to academic credentials while prioritizing user privacy. Integrating.

the system with academic institutions introduces another layer of complexity, as it involves overcoming challenges associated with diverse data.

formats, verification processes, and technological infrastructures across different institutions. Furthermore, the implementation must address issues related to hashing algorithms and data integrity to ensure the accuracy and consistency of academic credential records on the blockchain. Tackling these implementation challenges requires a meticulous approach to system architecture, security measures, and seamless integration with various stakeholders in the educational ecosystem.

The implementation of blockchain technology in educational verifier credentials offers a promising solution to various challenges entrenched within conventional credential verification systems. One of the foremost issues tackled is the susceptibility to forgery and fraud inherent in paper-based certificates. Blockchain's immutable ledger ensures that once a credential is recorded, it cannot be retroactively altered or tampered with, thereby guaranteeing its authenticity. By employing cryptographic signatures unique to each credential, blockchain adds an additional layer of security, thwarting attempts at counterfeiting. Moreover, the decentralized nature of blockchain transforms the verification process, shifting away from reliance on centralized authorities. This decentralization not only expedites verification but also diminishes the potential for delays and inefficiencies commonly associated with traditional systems.

Furthermore, blockchain technology addresses privacy concerns by enabling the implementation of self-sovereign identity frameworks. Individuals retain full control over their credentials, allowing them to selectively disclose information as needed. This empowers individuals with greater privacy and data security, mitigating risks associated with sharing sensitive educational credentials. Additionally, blockchain streamlines th e verification process, offering instantaneous verification that significantly reduces verification delays and associated costs. Its standardized platform enhances the global recognition of credentials, bridging the gaps created by differences in educational systems and standards across borders. Despite these advancements, challenges such as scalability, interoperability, and regulatory compliance persist, necessitating continued refinement and innovation in the adoption of blockchain-based educational verifier credentials. Nonetheless, the transformative potential of blockchain holds promise in revolutionizing credential verification,offering a more secure, efficient, and universally recognized solution.

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APPENDIX

A. SOURCE CODE

```
{
  "name": "fabcar",
  "version": "0.0.1",
  "description": "e-store contract",
  "main": "index.js",
  "engines":      {
    "node": ">=8",
    "npm": ">=5"
  },
  "scripts": {
    "lint":     "eslint     .",
    "pretest": "npm run lint",
    "test": "nyc mocha test --recursive",
    "start": "fabric-chaincode-node start",
    "mocha": "mocha test --recursive"
  },
  "engineStrict": true,
  "author":   "emali",
  "dependencies": {
    "crypto-js": "latest",
    "fabric-contract-api": "^2.1.2",
    "fabric-shim": "^2.1.2",
    "jsrsasign": "^8.0.20",
    "merkletreejs": "^0.2.8"
  },
  "devDependencies": {
    "chai": "^4.1.2",
    "chai-as-promised": "^7.1.1",
    "eslint": "^5.16.0",
    "mocha": "^7.2.0",
```

```
    "nyc": "^15.1.0",
    "sinon": "^7.1.1",
    "sinon-chai": "^3.2.0"
  },
  "nyc": {
    "exclude":     [
      "coverage/**",
      "test/**"
    ],
    "reporter": [
      "text-summary",
      "html"
    ],
    "all": true,
    "check-coverage": true,
    "statements": 100,
    "branches": 100,
    "funtions": 100,
    "lines": 100
  }
}
{
  "name": "fabcar",
  "version": "0.0.1",
  "description": "e-store contract",
  "main": "index.js",
  "engines":     {
    "node": ">=8",
    "npm": ">=5"
  },
  "scripts": {
    "lint":     "eslint     .",
    "pretest": "npm run lint",
    "test": "nyc mocha test --recursive",
```

```
    "start": "fabric-chaincode-node start",
    "mocha": "mocha test --recursive"
},
"engineStrict": true,
"author":     "emali",
"dependencies": {
    "crypto-js": "latest",
    "fabric-contract-api": "^2.1.2",
    "fabric-shim": "^2.1.2",
    "jsrsasign": "^8.0.20",
    "merkletreejs": "^0.2.8"
},
"devDependencies": {
    "chai": "^4.1.2",
    "chai-as-promised": "^7.1.1",
    "eslint": "^5.16.0",
    "mocha": "^7.2.0",
    "nyc": "^15.1.0",
    "sinon": "^7.1.1",
    "sinon-chai": "^3.2.0"
},
"nyc": {
    "exclude":        [
        "coverage/**",
        "test/**"
    ],
    "reporter": [
        "text-summary",
        "html"
    ],
    "all": true,
    "check-coverage": true,
    "statements": 100,
    "branches": 100,
```

```
        "funtions": 100,
        "lines": 100
    }
}
let logger = require("../services/logger");
let encryption = require('../services/encryption');
let certificates = require('../database/models/certificates');



async function getGenerateProof(req,res,next) {

    try {

        if (!req.query.sharedAttributes || req.query.sharedAttributes.length === 0) {throw
            Error("Choose atleast one attribute to share")
        }

        let mTreeProof = await
encryption.generateCertificateProof(req.query.sharedAttributes, req.query.certUUID,
req.session.email);
        let disclosedData = await certificates.findOne({"_id" :
req.query.certUUID}).select(req.query.sharedAttributes.join(" ") + " -_id");

        res.status(200).send({
            proof: mTreeProof,
            disclosedData: disclosedData,
            certUUID: req.query.certUUID
        })
    } catch (e) {
        logger.error(e);
        next(e);
    }
}
```

```
async function apiErrorHandler(err, req, res, next) {

    // set locals, only providing error in development
    res.locals.message = err.message;
    res.locals.error = req.app.get('env') === 'development' ? err : {};

    // render the error page
    return res.status(err.status || 500).send(JSON.stringify(err.message, undefined,
4));
}


module.exports = {getGenerateProof, apiErrorHandler};
let students = require('../database/models/students');
let fabricEnrollment = require('../services/fabric/enrollment');
let chaincode = require('../services/fabric/chaincode'); let
logger = require("../services/logger");
let studentService = require('../services/student-service');

let title = "Student Dashboard";
let root = "student";


async function postRegisterStudent(req, res, next) {
    try {
        let keys = await fabricEnrollment.registerUser(req.body.email);

        let dbResponse = await students.create({
            name : req.body.name,
            email:          req.body.email,
            password: req.body.password,
            publicKey: keys.publicKey
```

```
    });

    res.render("register-success",  {  title,  root,
      logInType: req.session.user_type || "none"});
  }
  catch    (e)    {
    logger.error(e);
    next(e);
  }
}

async function logOutAndRedirect (req, res, next) {
  req.session.destroy(function () {
    res.redirect('/');
  });
};


async function postLoginStudent (req,res,next)
  { try {
    let    studentObject    =    await    students.validateByCredentials(req.body.email,
req.body.password)

    req.session.user_id     =     studentObject._id;
    req.session.user_type        =        "student";
    req.session.email      =      studentObject.email;
    req.session.name      =      studentObject.name;
    req.session.publicKey = studentObject.publicKey;

    return res.redirect("/student/dashboard")
  }  catch  (e)  {
    logger.error(e);
    next(e);
```

```
    }
}


async function getDashboard(req, res, next) {
    try {
        let                          certData                          =                          await
studentService.getCertificateDataforDashboard(req.session.publicKey,
req.session.email);
        res.render("dashboard-student", { title, root, certData,
            logInType: req.session.user_type || "none"});

    }   catch   (e)   {
        logger.error(e);
        next(e);
    }
}

module.exports     =     {postRegisterStudent,     postLoginStudent,     logOutAndRedirect,
getDashboard};
let  universities  =  require('../database/models/universities');
let fabricEnrollment = require('../services/fabric/enrollment');
let  chaincode  =  require('../services/fabric/chaincode');  let
logger = require("../services/logger");
let universityService = require("../services/university-service");


let title = "University";
let root = "university";


async function postRegisterUniversity(req, res, next)
    { try {
        let keys = await fabricEnrollment.registerUser(req.body.email);
```

```
        let location = req.body.location + `, ${req.body.country}`;

        let dbResponse = await universities.create({
            name : req.body.name,
            email:             req.body.email,
            description: req.body.description,
            location: location,
            password: req.body.password,
            publicKey: keys.publicKey
        });

        let result = await chaincode.invokeChaincode("registerUniversity",
            [    req.body.name,    keys.publicKey,    location,    req.body.description],    false,
req.body.email);
        logger.debug(`University Registered. Ledger profile: ${result}`);

        res.render("register-success",   {   title,   root,
            logInType: req.session.user_type || "none"});
    }
    catch      (e)      {
        logger.error(e);
        next(e);
    }
}

async function postLoginUniversity (req,res,next) {
    try {
        let    universityObject    =    await    universities.validateByCredentials(req.body.email,
req.body.password)
        req.session.user_id = universityObject._id;
        req.session.user_type     =     "university";
        req.session.email = universityObject.email;
        req.session.name = universityObject.name;
```

```
        return res.redirect("/university/issue")
    }   catch   (e)   {
        logger.error(e);
        next(e);
    }
}

async function logOutAndRedirect (req, res, next) {
    req.session.destroy(function () {
        res.redirect('/');
    });
}

async function postIssueCertificate(req,res,next) {
    try {
        let certData = {
            studentEmail:     req.body.studentEmail,
            studentName:     req.body.studentName,
            universityName:        req.session.name,
            universityEmail:        req.session.email,
            major:                          req.body.major,
            departmentName:  req.body.department,
            cgpa: req.body.cgpa,
            dateOfIssuing: req.body.date,
        };

        let serviceResponse = await universityService.issueCertificate(certData);

        if(serviceResponse) {
            res.render("issue-success",    {    title,    root,
                logInType: req.session.user_type || "none"});
        }

    } catch (e) {
```

```
        logger.error(e);
        next(e);
    }
}

async function getDashboard(req, res, next) {
    try {
        let certData = await
universityService.getCertificateDataforDashboard(req.session.name,
req.session.email);
        res.render("dashboard-university", { title, root, certData,
            logInType: req.session.user_type || "none"});

    } catch (e) {
        logger.error(e);
        next(e);
    }
}
module.exports = {postRegisterUniversity, postLoginUniversity, logOutAndRedirect,
postIssueCertificate, getDashboard};
let logger = require("../services/logger");
let encryption = require('../services/encryption');
let certificates =
require('../database/models/certificates'); let moment =
require('moment'); let title = "Verification Portal";
let root = "verify";
async function postVerify(req,res,next) {
    try {
        let proofObject = req.body.proofObject;
        proofObject = JSON.parse(proofObject);

        if (!proofObject.disclosedData || Object.keys(proofObject.disclosedData).length
===0 ){
throw new Error("No parameter given. Provide parameters that need to be
```

```
verified");
        }
        let proofIsCorrect = await encryption.verifyCertificateProof(proofObject.proof,
proofObject.disclosedData, proofObject.certUUID );

        if (proofIsCorrect) {
            let certificateDbObject = await certificates.findOne({"_id":
proofObject.certUUID}).select("studentName studentEmail _id dateOfIssuing
universityName universityEmail");

            res.render("verify-success", { title, root,
                logInType: req.session.user_type || "none",
                certData : certificateDbObject, proofData :
                proofObject.disclosedData
            })

        } else {
            res.render("verify-fail", {
                title, root,
                logInType: req.session.user_type || "none"
            })
        }

    } catch (e) {
        logger.error(e);
        next(e);
    }
}

module.exports = {postVerify};

//initialize env variables, database and loaders.
const config = require('./loaders/config');
```

```
//load database
const mongoose = require('./database/mongoose');

//load fabric environemtn
require('./loaders/fabric-loader');



//third party libraries
let createError = require('http-errors');
let express = require('express');
let path = require('path');
let cookieParser = require('cookie-parser');
let morgan = require('morgan'); const
helmet = require('helmet');
const cors = require('cors');

//local imports
let limiter = require('./middleware/rate-limiter-
middleware'); const logger = require('./services/logger');
const sessionMiddleware = require('./loaders/express-session-loader');

//Router imports
let indexRouter = require('./routes/index-router');
let apiRouter = require('./routes/api-router');
let universityRouter = require('./routes/university-router');
let studentRouter = require('./routes/student-router'); let
verifyRouter = require('./routes/verify-router');

//express
let app = express();

// view engine setup
app.set('views', path.join(__dirname,
'views')); app.set('view engine', 'ejs');
```

```
//middleware

app.use(cors());
app.use(limiter.rateLimiterMiddlewareInMemory);
app.use(morgan('tiny', { stream: logger.stream }));
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(cookieParser());
app.use(express.static(path.join(__dirname, 'public')));
app.use(helmet());

app.use(sessionMiddleware);

//routers
app.use('/',            indexRouter);
app.use('/api',         apiRouter);
app.use('/university', universityRouter);
app.use('/student',     studentRouter);
app.use('/verify', verifyRouter);
// catch 404 and forward to error handler
app.use(function(req, res, next) {
  next(createError(404));
});

// error handler
app.use(function(err, req, res, next) {
  // set locals, only providing error in development
  res.locals.message = err.message;
  res.locals.error = req.app.get('env') === 'development' ? err : {};

  // render the error page
  res.status(err.status || 500);
  res.render('error');

   })
```